

Skill 1

```
mysql> ALTER TABLE bank_transactions
-> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESCRIBE bank_transactions;
ERROR 1146 (42S02): Table 'skill1_bank.bank_transactions' doesn't exist
mysql> DESCRIBE bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(50)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

7 rows in set (0.00 sec)

```
mysql> ALTER TABLE bank_transactions
-> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESCRIBE bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(100)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

7 rows in set (0.00 sec)

```
mysql> RENAME TABLE bank_transactions
    -> TO customer_transactions;
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_skill1_bank |
+-----+
| customer_transactions |
+-----+
1 row in set (0.01 sec)

mysql> TRUNCATE TABLE customer_transactions;
Query OK, 0 rows affected (0.04 sec)

mysql> RENAME TABLE customer_transactions
    -> TO bank_transactions;
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_skill1_bank |
+-----+
| bank_transactions      |
+-----+
1 row in set (0.00 sec)

mysql> CREATE TABLE bank_backup (
    ->     id INT,
    ->     data VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> DROP TABLE bank_backup;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SHOW TABLES;
+-----+
| Tables_in_skill1_bank |
+-----+
| bank_transactions      |
+-----+
1 row in set (0.00 sec)

mysql> INSERT INTO bank_transactions VALUES
    -> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
    -> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
    -> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
    -> (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
    -> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
    -> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
    -> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
    -> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
    -> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
    -> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
mysql> INSERT INTO bank_transactions
    -> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
    -> VALUES
    -> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
    -> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
    -> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
    -> (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
    -> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
    -> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
    -> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
    -> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
    -> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
    -> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM bank_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

```
10 rows in set (0.00 sec)
```

```
mysql> INSERT INTO bank_transactions
-> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
-> VALUES
-> (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM bank_transactions
-> WHERE txn_id = 111;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
111	Venu	Vizag	Deposit	7000.00	2024-01-18	NULL

```
1 row in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> WHERE txn_id = 105;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL

```
1 row in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> WHERE txn_id = 105;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL

```
1 row in set (0.00 sec)
```

```
mysql> DELETE FROM bank_transactions
-> WHERE txn_id = 111;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> WHERE txn_id = 111;
```

```
Empty set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
--------	---------------	-------------	------------------	--------	------------------	------------

```
mysql> SELECT *
-> FROM bank_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> WHERE transaction_type = 'Deposit';
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> ORDER BY amount DESC;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL

```
10 rows in set (0.00 sec)
```

```
mysql> GRANT SELECT
-> ON bank_transactions
-> TO Auditor1;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> GRANT ALL
-> ON bank_transactions
-> TO BranchManager;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> REVOKE SELECT
-> ON bank_transactions
-> FROM Auditor1;
ERROR 1147 (42000): There is no such grant defined for user 'Auditor1' on host '%' on table 'bank_transactions'
mysql> REVOKE ALL
-> ON bank_transactions
-> FROM BranchManager;
ERROR 1147 (42000): There is no such grant defined for user 'BranchManager' on host '%' on table 'bank_transactions'
mysql>
```

```

-> TO Auditor1;
ERROR 1419 (42000): You are not allowed to create a user with GRANT
mysql> GRANT ALL
-> ON bank_transactions
-> TO BranchManager;
ERROR 1419 (42000): You are not allowed to create a user with GRANT
mysql> REVOKE SELECT
-> ON bank_transactions
-> FROM Auditor1;
ERROR 1147 (42000): There is no such grant defined for user 'Auditor1' on host '%' on table 'bank_transactions'
mysql> REVOKE ALL
-> ON bank_transactions
-> FROM BranchManager;
ERROR 1147 (42000): There is no such grant defined for user 'BranchManager' on host '%' on table 'bank_transactions'
mysql> REVOKE ALL
-> ON bank_transactions
-> FROM BranchManager;
ERROR 1147 (42000): There is no such grant defined for user 'BranchManager' on host '%' on table 'bank_transactions'
mysql> UPDATE bank_transactions
-> SET amount = 6000
-> WHERE txn_id = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
-> SET amount = 99999
-> WHERE txn_id = 102;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> ROLLBACK TO Before_Update;
ERROR 1305 (42000): SAVEPOINT Before_Update does not exist
mysql> SELECT txn_id, customer_name, amount
-> FROM bank_transactions
-> WHERE txn_id = 102;

```

```
mysql> SELECT txn_id, customer_name, amount
-> FROM bank_transactions
-> WHERE txn_id = 102;
```

```
+-----+-----+-----+
| txn_id | customer_name | amount |
+-----+-----+-----+
|    102 | Sita          | 99999.00 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> UPDATE bank_transactions
-> SET amount = 7000
-> WHERE txn_id = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> UPDATE bank_transactions
-> SET amount = 9000
-> WHERE txn_id = 102;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT txn_id, customer_name, amount
-> FROM bank_transactions
-> WHERE txn_id IN (101,102);
```

```
+-----+-----+-----+
| txn_id | customer_name | amount |
+-----+-----+-----+
|    101 | Ravi          | 7000.00 |
|    102 | Sita          | 9000.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```